Programming for AI 1

PHIL 121 (soon to be HCAI 121)

Eric Pacuit

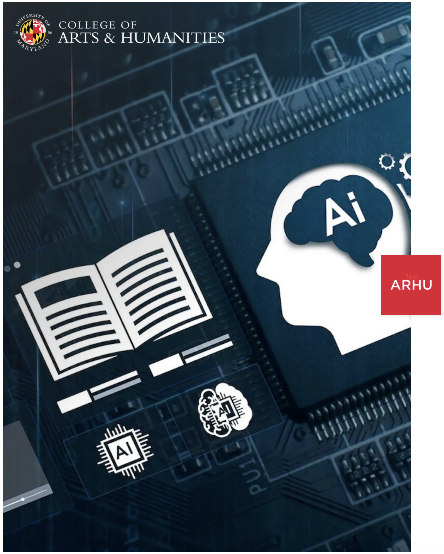
Department of Philosophy
University of Maryland

August 31, 2026

Course Information

Instructor:	Eric Pacuit Email: epacuit@umd.edu Website: pacuit.org Office: Skinner 1103A
Office Hours:	TBA
Class Times:	MW 2:00pm - 3:15pm
Course Website:	`https://umd.instructure.com/courses/1411542`

AI at UMD



APPLY GIVE MENU

COLLEGE OF ARTS AND HUMANITIES, PHILOSOPHY, THE HARRIET TUBMAN DEPARTMENT OF WOMEN, GENDER, AND SEXUALITY STUDIES

UMD to Launch 2 Bachelor's Degrees to Explore Technical, Human Dimensions of AI

New majors aim to ethically advance AI tech, prepare students to address its societal impact.

▶ *Read More*

BA in Human-Centered AI (HCAI)

An interdisciplinary major that prepares students not only to adapt to AI, but to shape how it is used, governed, and understood.

BA in Human-Centered AI (HCAI)

An interdisciplinary major that prepares students not only to adapt to AI, but to shape how it is used, governed, and understood.

Two objectives:

1. **Technical fluency**: Make the technical foundations of AI accessible to a broad range of students, so they can meaningfully engage with AI systems.
2. **Disciplinary perspective**: Train students in human-facing disciplines, so they can understand, evaluate, and shape AI systems.

https://philosophy.umd.edu/academic-programs/undergraduate-program/bachelor-arts-human-centered-ai

HCAI Courses

Technical core

HCAI 121
Programming for AI 1

HCAI 122
Programming for AI 2

HCAI 200
Formal Methods in AI

HCAI 300
Artificial Intelligence

HCAI 301
Machine Learning

HCAI Courses

Technical core	Ethics & social core
HCAI 121 Programming for AI 1	HCAI 101 AI & The Human Experience
HCAI 122 Programming for AI 2	PHIL 211 Ethics and AI
HCAI 200 Formal Methods in AI	WGSS 115 Gender, Race & Computing
HCAI 300 Artificial Intelligence	INST 201 Designing Fair Systems
HCAI 301 Machine Learning	HCAI 490 Capstone

HCAI Courses

Technical core	Ethics & social core	Specialization
HCAI 121 Programming for AI 1	HCAI 101 AI & The Human Experience	Arts
HCAI 122 Programming for AI 2	PHIL 211 Ethics and AI	Design & UX
HCAI 200 Formal Methods in AI	WGSS 115 Gender, Race & Computing	Ethics
HCAI 300 Artificial Intelligence	INST 201 Designing Fair Systems	Language & Cognition
HCAI 301 Machine Learning	HCAI 490 Capstone	Law, Policy & Governance
		Logic, Epistemology & ML
		Society, Culture & Tech.
		More to be added. . .

HCAI Courses

Technical core	Ethics & social core	Specialization
HCAI 121 Programming for AI 1	HCAI 101 AI & The Human Experience	Arts
HCAI 122 Programming for AI 2	PHIL 211 Ethics and AI	Design & UX
HCAI 200 Formal Methods in AI	WGSS 115 Gender, Race & Computing	Ethics
HCAI 300 Artificial Intelligence	INST 201 Designing Fair Systems	Language & Cognition
HCAI 301 Machine Learning	HCAI 490 Capstone	Law, Policy & Governance
		Logic, Epistemology & ML
		Society, Culture & Tech.
		More to be added...

Online Resources

- **ELMS Course Website**:
  https://umd.instructure.com/courses/1411542
- **PollEverywhere**: https://PollEv.com/epacuit
  Make sure to use your terpmail address when logging in so that your answers can be added to your participation grade.
- **Piazza**:
  umd.instructure.com/courses/1411542/external_tools/90732

Prerequisites

This course is designed for students with **little to no prior programming experience** to build a solid foundation in Python programming and the problem-solving skills essential for working with AI.

Grading

Participation	10%
Weekly Quizzes	30%
Programming Projects	30%
Midterm	15%
Final Exam	15%

Participation

Attending lectures is essential for success in this course.

You are responsible for all material covered during lectures, even if you are absent. It will be very easy to fall behind if you miss classes, so please reach out if you miss more than one or two sessions to ensure you stay on track.

Participation

Attending lectures is essential for success in this course.

You are responsible for all material covered during lectures, even if you are absent. It will be very easy to fall behind if you miss classes, so please reach out if you miss more than one or two sessions to ensure you stay on track.

Your participation grade will be based on two scores:

- Periodic PollEverywhere responses (usually survey questions)
- Weekly question/comment: Each week, you must submit a question or comment by **Tuesday at 11pm**.

Weekly Quizzes

Every Wednesday, there will be a short in-person quiz covering material introduced on Monday.

- The first quiz will be **Wednesday, September 9**.
- Each quiz will be multiple choice/free response and last 15-20 minutes.
- The quiz will be given at the beginning of class. **Please do not be late**.
- If you miss a quiz, the make-up must be scheduled within 1 week.

Programming Projects

There will be two main programming projects, and a number of smaller projects assigned periodically throughout the semester. More details about these projects will be provided later in the semester.

Exams

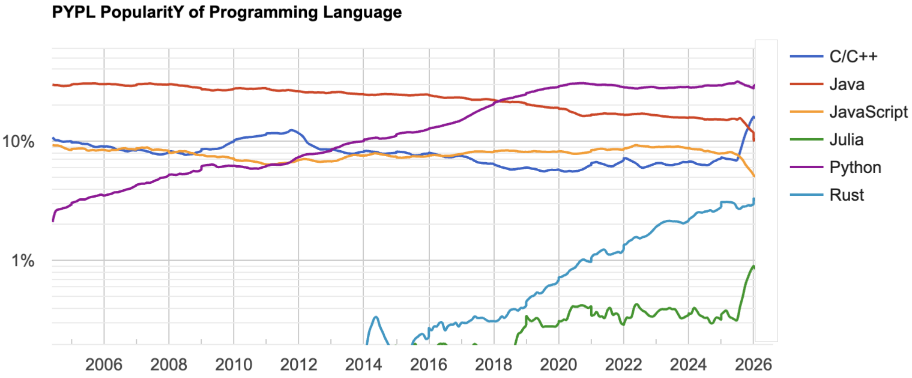
- Midterm Exam: in-person exam given in class during week 7.
- Final Exam: in-person exam given during exam week.

Python is a high-level, interpreted, general-purpose programming language known for its simplicity, readability, and versatility. Created by Guido van Rossum and first released in 1991, Python is widely used for various applications, especially Machine Learning and Data Science.

www.python.org

python™

Python is a very popular language!



https://pypl.github.io/PYPL.html

Python Libraries

Scientific Computing, Data Science, Machine Learning Ecosystem

Working with data

NumPy	numerical arrays	https://numpy.org/
pandas	data tables	https://pandas.pydata.org/
SciPy	scientific computing	https://scipy.org/

Visualizing data

Matplotlib	plots and charts	https://matplotlib.org/
Seaborn	statistical graphics	https://seaborn.pydata.org/

Python Libraries

Scientific Computing, Data Science, Machine Learning Ecosystem

Working with the web

Requests	fetch web pages & APIs https://requests.readthedocs.io/
BeautifulSoup	read data from web pages https://www.crummy.com/software/BeautifulSoup/
FastAPI	build web services https://fastapi.tiangolo.com/

Python Libraries

Scientific Computing, Data Science, Machine Learning Ecosystem

Machine learning

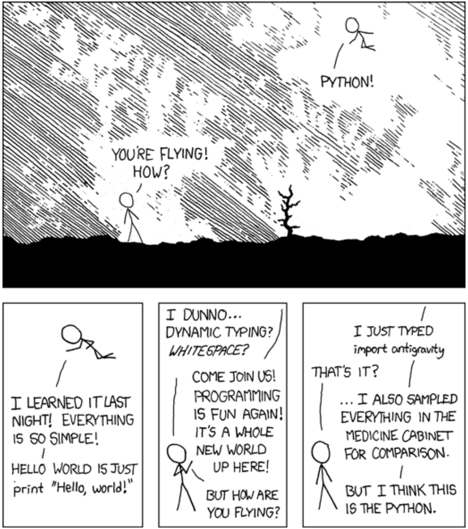
scikit-learn	classical ML	`https://scikit-learn.org/`
PyTorch	neural networks	`https://pytorch.org/`
TensorFlow/Keras	deep learning	`https://keras.io/`

Language & AI

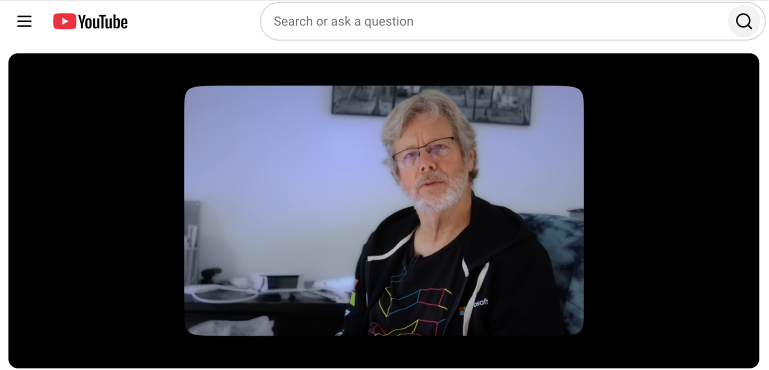
NLTK	text processing	`https://www.nltk.org/`
Ollama	run LLMs locally	`https://ollama.com/`

Plus many others...

Python is Fun!



The Story of Python

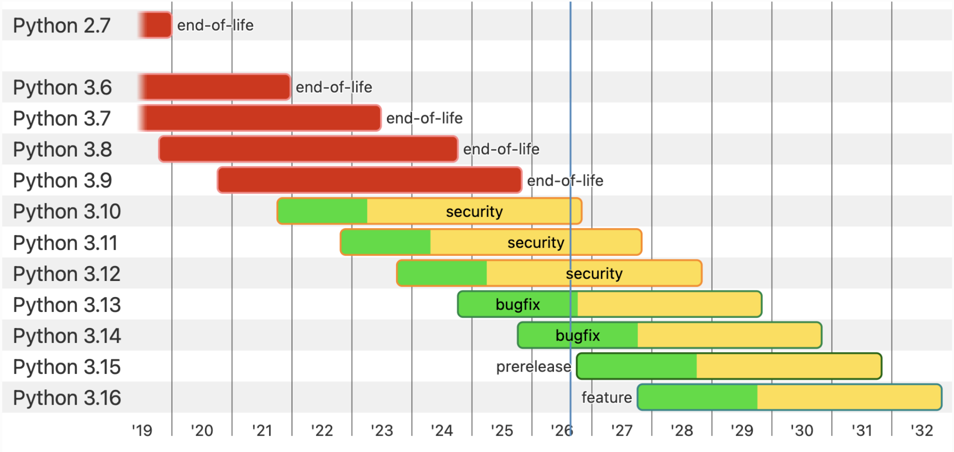


The Story of Python and how it took over the world | Python: The Documentary

https://www.youtube.com/watch?v=GfH4QL4VqJ0

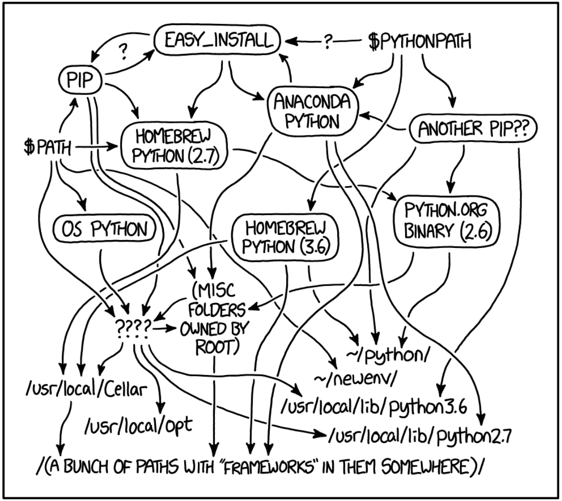
(This might be more interesting to watch at the end of the semester...)

Installing Python



https://www.python.org/downloads/

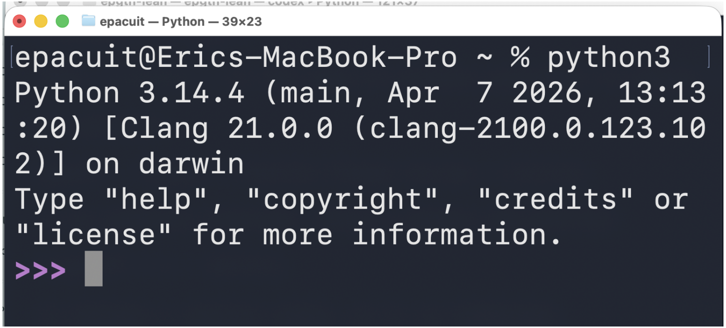
Installing Python



MY PYTHON ENVIRONMENT HAS BECOME SO DEGRADED THAT MY LAPTOP HAS BEEN DECLARED A SUPERFUND SITE.

Installing Python

Python might already be installed on your computer (especially if you have a Mac). Open a terminal and type `python` or `python3`.



Installing Python

Recommendation:
(This is not needed for today, but it is a good idea to set up your local environment)

1. On a Mac, make sure the ...
2. Install VS Code
   `https://code.visualstudio.com/`
3. Install the Python extension for VS Code
   `https://marketplace.visualstudio.com/items?itemName=ms-python.python`
4. Install the uv Python package manager
   `https://docs.astral.sh/uv/getting-started/installation/`
   - Use `uv` to run Python scripts and to install Python
     `https://docs.astral.sh/uv/getting-started/features/`

Jupyter and Colab

To start, we'll write and run Python in the browser (so nothing to install).

- **Jupyter notebooks**: documents that mix code, results, and notes, where you run code one cell at a time.
  `https://jupyter.org/`
- **Google Colab**: runs Jupyter notebooks in your browser, free, with nothing to set up.
  `https://colab.research.google.com/`

You only need a Google account (your UMD account works).

Python is Interpreted

You type code and it runs immediately, line by line, no separate "build" step.

- Great for learning: try something, see what happens, adjust.
- This is in contrast to *compiled languages* like Rust, C/C++ or Java that translate the whole program first, then run it.

GitHub

Where we will keep and share code this semester.

- **GitHub**: a widely used online tool for storing and sharing code. https://github.com/

- You will use it to get assignments and turn in your work.
- We will learn about version control to track changes to code (**Git**: https://git-scm.com/) later in the course. For now, you just need an account.

To do this week: create a free GitHub account.
(Consider using your UMD email since you can apply for free student benefits.)

The New York Times Magazine

Account

Coding After Coders: The End of Computer Programming as We Know It

In the era of A.I. agents, many Silicon Valley programmers are now barely programming. Instead, what they're doing is deeply, deeply weird.

Programming in the Age of AI

You have probably noticed that AI tools can produce code.

- ChatGPT (Codex), Claude (Code), Gemini (CLI), Copilot, and others: generate code from a description.
- So why learn to program at all?

Someone has to know whether the code is **correct**.

- AI writes code that *looks* correct (and is sometimes wrong).
- This course builds the judgment to evaluate code: to read it, test it, and know when to trust it.

Why Python for AI?

Python is one of the best languages for working with AI, both for people and for the models themselves.

- Most major AI tools are built for Python.
- Python has a huge collection of well-supported and vetted libraries.
- AI models are especially strong at writing Python:
  - they were trained on an enormous amount of Python code,
  - they can run and check the Python they write, and
  - they are incredibly good at finding the right library and explaining how to use it.

A First Look: Wordle

Let's test this idea.

- We asked several AI tools to write the game **Wordle** in Python.
- Each produced code that seems to work.
- Our job is to read it, run it, and decide whether it is actually *correct*.

You will understand every line of the code that is produced by the models by the end of the course.